

[DC] Designing Immersive Tools for Supporting Cognition in Remote Scientific Collaboration

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ABSTRACT

Scientists collaborate remotely across institutions, countries and continents. However, collaborating remotely is challenging. Video-conferencing tools used for meetings limit the cognitive practices that collaborators can partake in. In virtual reality (VR) users can gain back spatial affordances present in collocated collaboration and we can design interactions that would not be possible in the real world. My research aims to investigate how VR can support cognition in remote scientific collaboration through the design, development, and study of Embodied Notes: a cognitive support tool designed to be used in a collaborative virtual environment.

Index Terms: Human-centered computing—Collaborative and social computing systems and tools; Human-centered computing—Virtual reality

1 INTRODUCTION

Research in remote scientific collaboration has explored the use of audio and video conferencing to make collaboration across distances easier and more productive. However, remote collaboration poses several challenges to a team: awareness of colleagues and their context, establishing trust is difficult, and there is a lack of motivating sense of presence with others [7]. Additionally, video-conferencing, the standard for synchronous meetings in scientific research, limits the kind of cognitive offloading that can be done in a scientific group setting compared to a collocated scientific group meeting with access to physical whiteboards, sticky notes, analog note-taking (notebooks), and space. I hypothesize that virtual reality (VR) could provide affordances for supporting cognition in a scientific group setting that video-conferencing does not or cannot provide and propose ways we can use VR affordances to provide cognitive support in remote scientific collaboration.

In her book “Extended Mind” [8], Annie Murphy Paul synthesizes findings from the fields of embodied, situated, and distributed cognition to propose a framework for “thinking outside the brain”. Her framework draws upon Clark and Chalmer’s seminal article titled “The Extended Mind” [2] to define thinking outside the brain as “skillfully engaging entities external to our heads—the feelings and movements of our bodies, the physical spaces in which we learn and work, and the minds of the other people around us—drawing them into our own mental processes.” Paul specified nine principles for extending the mind: 1) offload information into the world; 2) transform information into an artifact and then interact with it; 3) seek to productively alter one’s state for improving mental labor; 4) re-embodiment information; 5) re-spatialize information; 6) re-socialize information; 7) generating cognitive loops; 8) thinking by creating cognitively congenial situations; and 9) thinking by embedding extensions in our everyday environments. In VR, scientists can more readily engage in these extended mind practices taking advantage of the spatial and embodied interactions facilitated by VR appli-



Figure 1: Scientists in needfinding study meeting in Spatial to discuss research over 4 weeks

cations to absorb complicated material and collaborate to develop new insights. However, more research is needed to examine how to leverage features and affordances available in VR to enhance cognition in remote scientific collaborations.

Research in immersive analytics explores the use of engaging, embodied analysis tools to support data understanding and decision making [5]. However, very few [4] immersive analytics tools include the ability to take notes despite note taking being a critical tool for cognitive offloading, an important aspect of conducting scientific research [6]. Other research in virtual reality has explored digital sticky notes for spatial ideation [3] and using peripherals like digital tablets [1] and smartphones [3] as input devices. However, these were limited to single user scenarios and did not explore what it might look like to create and share notes in a collaborative setting. My research seeks to fill this gap, by exploring the design and development of a cognitive support tool designed to be used in a collaborative virtual environment: Embodied Notes.

2 RESEARCH APPROACH

My research aims to investigate how scientists can use virtual reality to support cognition in a remote collaborative setting. Below I describe three studies (one of which I have completed) I plan to conduct towards this aim: a needfinding study of scientists using VR to discuss scientific research remotely, a user study of note taking and communication tool Embodied Notes, and an extension of Embodied Notes to capture shared artifact states.

2.1 Needfinding Study with Scientists

To identify the needs of scientists collaborating remotely I conducted a qualitative study where we observed two interdisciplinary groups of scientists ($n=14$) including professors, graduate students, and post-docs. Their research areas spanned computer science, marine biology, mathematics, computational biology, coral biology and biochemistry. Each group met in VR over a period of 4 weeks to discuss their research. They met using the meeting application Spatial where users are able to import multiple data formats including 3D models, pdfs, and images and create avatars that resemble themselves. One group also used Nanome for exploring biological data. During the meeting we recorded their interactions, and after each meeting we had participants complete a survey and a group interview to gain insight into their experiences with meeting in VR. We conducted a thematic analysis of transcripts from group discussions and open responses questionnaires to understand their data and self organization practices, identified affordances unique to VR for supporting cognition in a collaborative setting and generated design

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recommendations for future tools. Our investigation was guided by the following research questions: RQ1) How do scientists use the virtual environment to organize their data and themselves? RQ2) What affordances in VR provide cognitive support for scientific collaboration? RQ3) How can these affordances (RQ2) be expanded upon to provide additional cognitive support?

One of our findings related to note taking and communication indicated that the scientists wanted to be able to communicate with others through both private and public channels and felt frustrated with the inability to write personal notes or quickly write down a continuous stream of thoughts as one would on pen and paper or with a physical keyboard. For example, they wanted to ask a question but not interrupt the current flow of conversation, write a note for themselves to explore or extend an idea further, and to share a thought privately with another individual. In video meeting applications like Zoom, that scientists traditionally use, they can send private messages to each other and put a question in the chat. For in-person meetings they can send an email or use apps like Slack when their laptop or phone is in reach.

2.2 Embodied Notes for Thoughts

To address the need for continuous personal note taking I am designing and developing Embodied Notes a tool for capturing and physicalizing stream of thought notes while in a VR meeting space with others, which are private by default but shareable with others. With Embodied Notes, users can record a note using their voice (users are automatically muted while recording) and take pictures to provide context for the note. The notes are represented as thought bubbles (spheres) where the outer texture of the sphere represents the visual context of where the note was taken. Users can grab a note from their pocket and bring it up to their mouth like a microphone to start recording their thoughts. They can replay the note using a shaking gesture, delete the note by throwing it upwards to the sky, share the note with everyone by throwing it to the ground, share with a specific person by pointing and throwing it at them, and store them in a notebook made visible by making a palms up open book gesture. Important considerations for the design of Embodied Notes was ensuring privacy when recording notes and providing observable cues for others during note taking. I plan to conduct a user study with Embodied Notes to answer the following research questions: RQ4) How does Embodied Notes support cognition during a scientific collaboration meeting in VR? RQ5) How do visual indicators of note taking affect collaborative discussions and group dynamics?



Figure 2: An Embodied Notes user holding a semi-transparent empty note and in the process of taking a picture of a coral before dictating a note (left). The final representation of a recorded note (right).

2.3 Capturing Shared Artifact States

Another finding from our needfinding study was participants wanted to be able to capture states of shared artifacts like whiteboards. In the desktop ecosystem, users can move content freely between

applications through mechanisms like copy and paste for text and images. I imagine a similar paradigm could be useful in and between VR applications. In the whiteboard scenario you might come up with an idea using the whiteboard while in a breakout group and then want to further refine your idea with the larger group outside the breakout room. One way one might do this is to take a picture, but there is power in being able to edit the work in its original form instead of working with a static image. As another example, a user could create a visualization in one application, then copy and paste a representation of that into a different application. For example, a scientist creates a visualization of a protein in one application and then brings that visualization into a different meeting application where they have their weekly group meeting. I intend to expand upon Embodied Notes to incorporate capturing the state of an object and its history. For example, placing an empty note on a whiteboard captures the markings that can then be placed on another whiteboard to be built upon at a later time or at a different whiteboard. With this extension I aim to answer the following research questions: RQ6) How can artifacts state be recorded or saved for continuity, reproducibility and sharing? RQ7) How do users use this feature?

3 CONCLUSION

I hypothesize that Embodied Notes will enable scientists to extend their thinking during remote scientific collaboration through offloading information into the virtual environment and enabling them to embody, spatialize, and socialize this information with themselves and others. Overall, my dissertation aims to make three contributions to the field of HCI: 1) provide a richer empirical understanding of how scientists use virtual environments for collaboration, 2) Embodied Notes for capturing, physicalizing and sharing stream of thought notes during a remote collaborative meeting in VR, 3) provide interaction techniques and design requirements for developing cognitive support tools for remote scientific collaboration based on findings from studying the use of Embodied Notes. Three main areas I would like feedback on are: 1) How can I improve the design of Embodied Notes? 2) Are there additional theories that could help me further ground my work? 3) Are there certain quantitative measures or experiments I should consider? What kind of tasks should I have users complete to ensure validity?

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